

MODEL TRAINING REPORT

Electronic Nose Coffee Classification System

Model: Multi-Layer Perceptron (MLP)

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TRAINING SUMMARY

Training Accuracy	Validation Accuracy	Training Time
100.0%	100.0%	2.1s

MODEL CONFIGURATION

Model Type	Multi-Layer Perceptron (MLP)
Epochs	150
Data Split	80% Training / 20% Validation
Architecture	Input -> Hidden Layers -> Sigmoid -> Output
Input Shape	8 sensors x 300 timesteps (flattened -> 2400)

GAS SENSORS USED (8 Sensors)

TGS2600	MQ135	MQ3	MQ6
MQ7	TGS2602	TGS2611	TGS2620

POWER CONSUMPTION DURING TRAINING

Power Metric	Value	Thermal / Energy	Value
Avg Total Power	7.24 W	Avg CPU Temp	47.5 °C
Peak Total Power	7.42 W	Peak CPU Temp	48.0 °C
Avg CPU+GPU Power	2.58 W	Energy Used	15.9 J (0.0044 Wh)
Peak CPU+GPU Power	2.79 W	Training Duration	2.1 s

MODEL EVALUATION RESULTS

Training Data — Accuracy: 100.00%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	1.0000	1.0000
Recall	1.0000	1.0000
F1-Score	1.0000	1.0000
Support (n samples)	96	96

Validation Data — Accuracy: 100.00%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	1.0000	1.0000
Recall	1.0000	1.0000
F1-Score	1.0000	1.0000
Support (n samples)	24	24

Confusion Matrix (Validation Data)

	Predicted: High	Predicted: Low
Actual: High	24	0
Actual: Low	0	24

TP = True Positive (High predicted as High)

FP = False Positive (Low predicted as High)

TN = True Negative (Low predicted as Low)

FN = False Negative (High predicted as Low)

MLP Summary -- 150 Epochs

Train Loss	0.027027
Validation Loss	0.020142
Total Epochs	150

Accuracy Curve (Checkpoint)

Epoch	Training Accuracy (%)	Validation Accuracy (%)
15	94.79	97.92
30	96.35	97.92
45	97.40	97.92
60	97.92	97.92
75	98.44	97.92
90	98.44	97.92
105	98.96	100.00
120	99.48	100.00
135	99.48	100.00
150	100.00	100.00

Loss Curve (Checkpoint)

Epoch	Training Loss	Validation Loss
15	0.123069	0.080637
30	0.094559	0.060043
45	0.078229	0.050432
60	0.065787	0.043140
75	0.055635	0.037344
90	0.047743	0.032170
105	0.041033	0.028544
120	0.035470	0.025561
135	0.030871	0.022861
150	0.027027	0.020142